

Sidecars in Kubernetes

Since you're learning **KCNA as a beginner**, let's understand **Sidecar Containers** in the easiest way.

1. What is a Sidecar?

A **sidecar container** is an additional container that runs **inside the same Pod** as the main application container.

Pod

|

├── Main Container

|

└── Application

|

└── Sidecar Container

 └── Helper task

★ Simple definition

Sidecar = A helper container that works alongside the main application container in the same Pod.

2. Real-Life Example 🏍️

Think about a **motorcycle with a sidecar**.

🏍️ Main motorcycle + 🚗 Sidecar

The motorcycle is the **main application**.

The sidecar helps the motorcycle but isn't the main vehicle.

Similarly:

Main Container + Sidecar Container

The sidecar performs a **supporting job** for the main container.

3. Why use a Sidecar?

A sidecar is used when your application needs an additional function.

For example:

-  Logging
 -  Security
 -  Monitoring
 -  Data synchronization
 -  Proxy/networking
 -  Configuration support
-

4. Example: Logging

Suppose your application creates logs:

Main Container

|

| creates logs



application.log

A sidecar can read those logs and send them somewhere else.

Pod

|

|— Main Container

| |

| |— Creates logs

|

|— Sidecar Container

|

└─ Reads logs



Log Server

The main application doesn't need to know how to send logs.

The sidecar handles it.

5. Main Container vs Sidecar

Main Container

Runs the main application

Main purpose of the Pod

Example: Web server

Performs business logic

Sidecar Container

Helps the main application

Supporting purpose

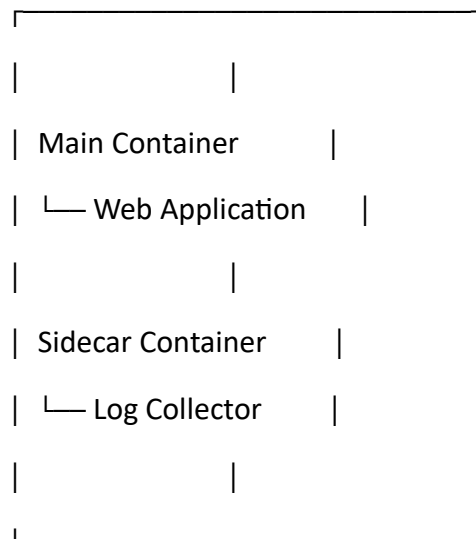
Example: Log collector

Performs additional tasks

6. Important: Same Pod ★ ★ ★

A sidecar runs **in the same Pod** as the main container.

Pod

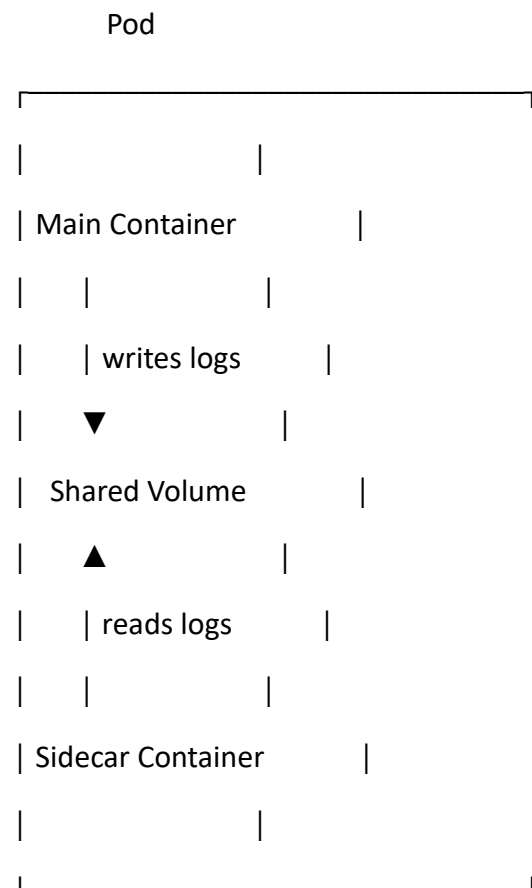


Because they are in the same Pod, containers can share things such as:

- Network
- IP address
- Volumes

7. Sidecar and Shared Volume

A very common example:



Both containers can access the same volume.

8. Sidecar vs Multiple Containers

Not every Pod with two containers necessarily means one is a sidecar.

For example:

Pod

└─ Nginx

└─ Application

These could be two containers working together.

A **sidecar** specifically means the additional container provides a **supporting function** to the main application.

9. Common Sidecar Examples

Logging Sidecar

Application



Logs



Sidecar



Log Server

Monitoring Sidecar

Application



Metrics



Sidecar



Monitoring System

Proxy Sidecar

Application



Sidecar Proxy



Network

A famous example is a **service-mesh proxy**, where a proxy container handles network-related tasks for the application.

10. Sidecar vs Init Container ★

This is important for Kubernetes exams.

Init Container

Runs **before** the main application container.

Init Container



Main Container

It prepares something for the application.

Sidecar

Runs alongside the main application.

Main Container



Sidecar Container

Easy memory:

Init = Before 🏠

Sidecar = Alongside ↔

🧠 KCNA Revision Notes

Sidecar

- A **sidecar** is a **helper container**.
- It runs in the **same Pod** as the main container.
- It supports the main application.
- It can share the **network** with the main container.
- It can share **volumes** with the main container.
- Common uses:
 - Logging
 - Monitoring
 - Security
 - Networking
 - Data synchronization
- Example:
- Main Container → Application

Sidecar → Log Collector

★ **One-line definition:**

A sidecar is a supporting container that runs alongside the main application container in the same Kubernetes Pod.

Easy memory trick:

 **Pod = House**

 **Main container = Person living in the house**

 **Sidecar = Helper living in the same house**

The helper doesn't replace the main person—it **supports them**.